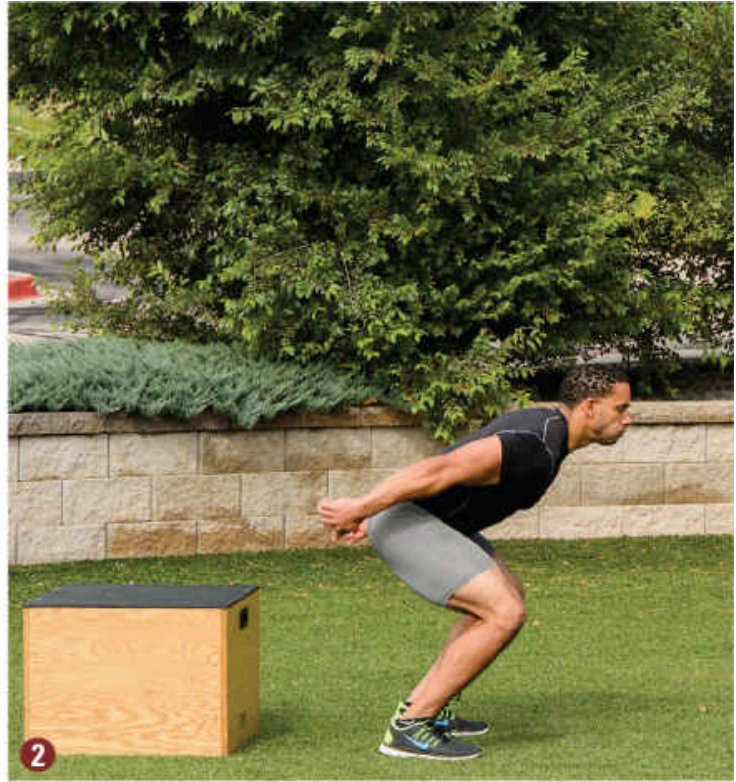


Upward movement: On landing, immediately jump forward as far as possible with both feet.

Note: When stepping from the box, step straight out. Do not first jump up or lower the center of gravity while stepping down, because these adjustments will change the height from which the exercise is performed. The time on the ground should be kept to a minimum; there is an optimal relationship between ground contact and resultant horizontal projection.

COMMON ERRORS

- **Not achieving an optimal forward-leaning angle through landing (i.e., too vertical and the ground contact time is increased or horizontal distance is lost, whereas leaning too far forward reduces the ground reaction force applied)**
- **Not pushing maximally into the ground to optimize the lower limb push for horizontal distance**



40

DEPTH JUMP TO 180-DEGREE TURN

Intensity level: High

Direction: Vertical and horizontal

Equipment: Plyometric box, 12 to 42 inches (30-107 cm) high

Starting position: Get into a comfortable, upright stance with feet hip-width apart on the plyometric box; toes should be near the edge of the box.

Arm action: Double arm

Preparatory movement: Step from the box.

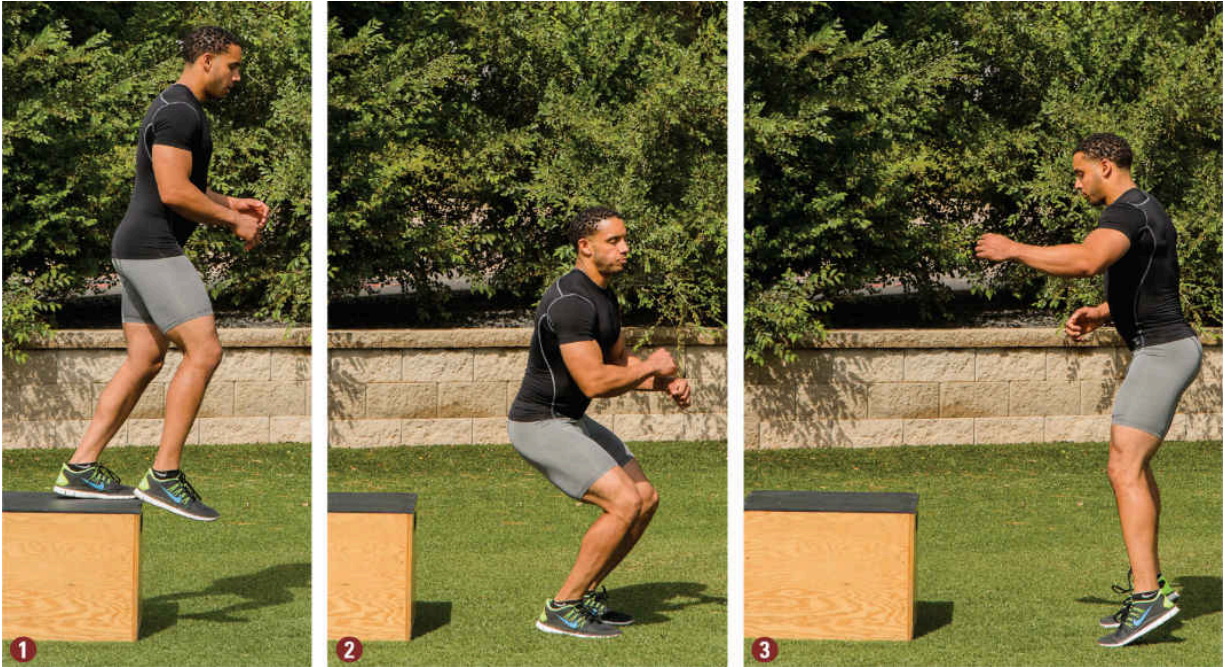
Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing with both feet.

Upward movement: On landing, immediately jump up as high as possible with both feet. While in the air, turn 180 degrees to land facing the opposite direction.

Note: When stepping from the box, step straight out. Do not first jump up or lower the center of gravity while stepping down, because these adjustments will change the height from which the exercise is performed. The time on the ground should be kept to a minimum.

COMMON ERRORS

- **Not controlling the rotational component of the rebound takeoff, leading to poor joint alignment in the 180-degree landing**
- **Not achieving sufficient ground contact to enable a rebound flight time that is long enough for a full turn to be completed**



41 DEPTH HOP

Intensity level: High

Direction: Vertical

Equipment: Plyometric box, 12 to 42 inches (30-107 cm) high

Starting position: Get into a comfortable, upright stance with feet hip-width apart on the plyometric box; toes should be near the edge of the box.

Arm action: Double arm

Preparatory movement: Step from the box with one foot.

Downward movement: Dorsiflex the ankles in preparation for an active flat-foot landing on one foot.

Upward movement: On landing, immediately hop up as high as possible with the landing foot. The other leg should be actively flexed during the upward movement to contribute to vertical momentum.

Note: When stepping from the box, step straight out. Do not first jump up or lower the center of gravity while stepping down, because these adjustments will change the height from which the exercise is

performed. The time on the ground should be kept to a minimum; there is an optimal relationship between ground contact time and resultant vertical and horizontal force. This is a very advanced form of the depth jump and should be performed only by those with adequate experience, ability, and strength as demonstrated in other versions of the depth jump.

COMMON ERRORS

- **Absorbing ground reaction force as the athlete lands through hip and knee flexion, which increases the amortization phase and ground contact time; the hips, knees, and ankles should be stiff when landing to optimize the ground contact time and rebound hop height**
- **Not dorsiflexing the ankle in preparation for ground contact; this negates the active flat foot contact and increases ground contact time**



UPPER BODY PLYOMETRIC DRILLS

Throws

42 CHEST PASS

Intensity level: Low

Direction: Forward

Equipment: Medicine or plyometric ball weighing 8 to 30 pounds (4-14 kg); wall or partner

Starting position: Get into a comfortable, upright stance with feet hip-width apart; face the wall or partner approximately 10 feet (3 m) away. Raise the ball to chest level with the elbows flexed.

Preparatory movement: Begin with a countermovement in the hips, knees, and ankles as well as the arms (i.e., cocking the arms by moving them slightly backward before the throw).

Arm action: Extend the legs forcefully, and transfer the power to both arms, throwing the ball to the wall or partner by extending the elbows. When the wall or partner returns the ball, catch it, return to the starting position, and immediately repeat the movement.

COMMON ERRORS

- **Not initiating the movement from the legs, reducing the impulse to the ball**
- **Not coordinating or sequencing the transfer of power from the lower body to the upper body, reducing the impulse to the ball**



43

TWO-HAND OVERHEAD THROW

Intensity level: Low

Direction: Forward and down

Equipment: Medicine or plyometric ball weighing 8 to 20 pounds (4-9 kg); wall or partner

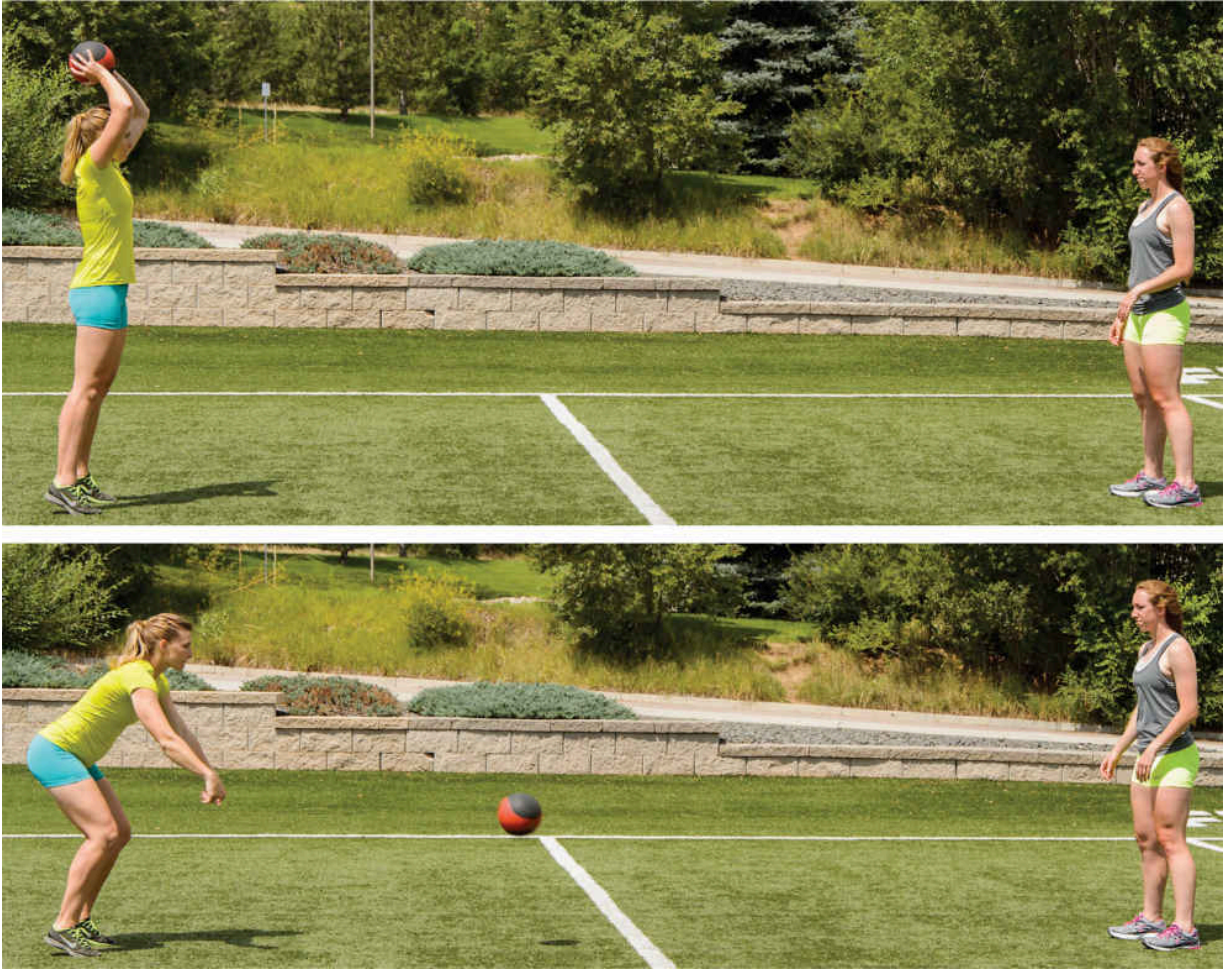
Starting position: Get into a comfortable, upright stance with feet hip-width apart; face the wall or partner approximately 10 feet (3 m) away. Raise the ball overhead.

Preparatory movement: Begin with a countermovement (i.e., cock the arms by moving them slightly backward before the throw).

Arm action: Using both arms, throw the ball to the wall or partner, keeping the elbows extended. When the wall or partner returns the ball, catch the ball and immediately repeat the throw. The partners can also bounce the ball on the ground between them for a downward movement and a catch on the rebound.

COMMON ERRORS

- **Arching the back in the start position, increasing lumbar spine involvement (rather than hip flexion) in the throwing action**
- **Not coordinating or sequencing downward throwing action so the optimal horizontal distance is not achieved**



44 ROTATIONAL THROW

Intensity level: Low

Direction: Forward through frontal plane

Equipment: Medicine or plyometric ball weighing 8 to 35 pounds (4-16 kg); wall

Starting position: Start in a lunge stance. The front leg (with the hip, knee, and ankle aligned) should face the wall approximately 3 to 6 feet (1-2 m) away. The rear leg should be externally rotated at the hip so that the foot is flat on the ground and pointing approximately parallel to the wall. Hold the ball at waist height in front of the body.

Preparatory movement: Begin with a countermovement so that the ball is brought back outside of the hip furthest from the wall and the

weight is shifted to this hip. (A countermovement for plyometric throws requires cocking the arms—that is, moving the arms slightly backward before the actual throw.)

Arm action: Initiate the movement with a rotational drive of the hips that continues through the trunk, bringing the ball across the body so that it gains momentum. Using both arms, complete the rotational arc of the ball so that it is released and thrown against the wall as it passes the front hip. Catch the rebound, and immediately repeat the movement.

Note: This exercise can be made more of a hip-initiated movement by shifting the weight onto the rear foot and abducting the hip (frontal plane) to bring weight onto the rear foot; this variation may benefit baseball pitchers seeking more leg-driven throwing action. Golfers may find it beneficial to rotate the hip in the transverse plane (not move in the frontal plane). Both variations enable the rear hip to initiate the action and accelerate the trunk rotation.

COMMON ERRORS

- **Rotating from the lumbar spine and trunk rather than shifting weight to facilitate the hip drive**
- **Over-relying on the upper body to throw the ball compared to using the lower body and hip rotation to enable a speed multiplication effect through the trunk to achieve greater impulse on the ball**



45 SINGLE-ARM THROW

Intensity level: Medium

Direction: Forward

Equipment: Medicine or plyometric ball weighing 1 to 5 pounds (0.5-2.3 kg); wall or partner

Starting position: Get into a comfortable, upright stance with feet hip-width apart; face the wall or partner approximately 10 feet (3 m) away. Raise the ball in one hand to a position of 90 degrees of shoulder abduction and 90 degrees of elbow flexion, with the arm rotated so the forearm is perpendicular to the ground.

Preparatory movement: Begin with a countermovement (i.e., cock the arm by moving it slightly backward before the throw).

Arm action: Using one arm, throw the ball to the wall or partner. When the wall or partner returns the ball, catch the ball and return to the starting position, allow the shoulder to externally rotate slightly, and immediately repeat the throw.

Note: This drill may also be performed using a natural throwing motion. Performing the action from a half-kneeling position may help isolate the upper body action of the throwing motion.

COMMON ERRORS

- **Not achieving a stable throwing position due to lack of thoracic mobility and overcompensations through lumbar spine flexion**
- **Not throwing from the shoulder but rotating through the trunk (lumbar spine); this is likely to occur when the ball is too heavy for the athlete**



46

POWER DROP

Intensity level: High

Direction: Upward

Equipment: Medicine or plyometric ball weighing 5 to 40 pounds (2-18 kg); partner; plyometric box 12 to 42 inches (30-107 cm) high

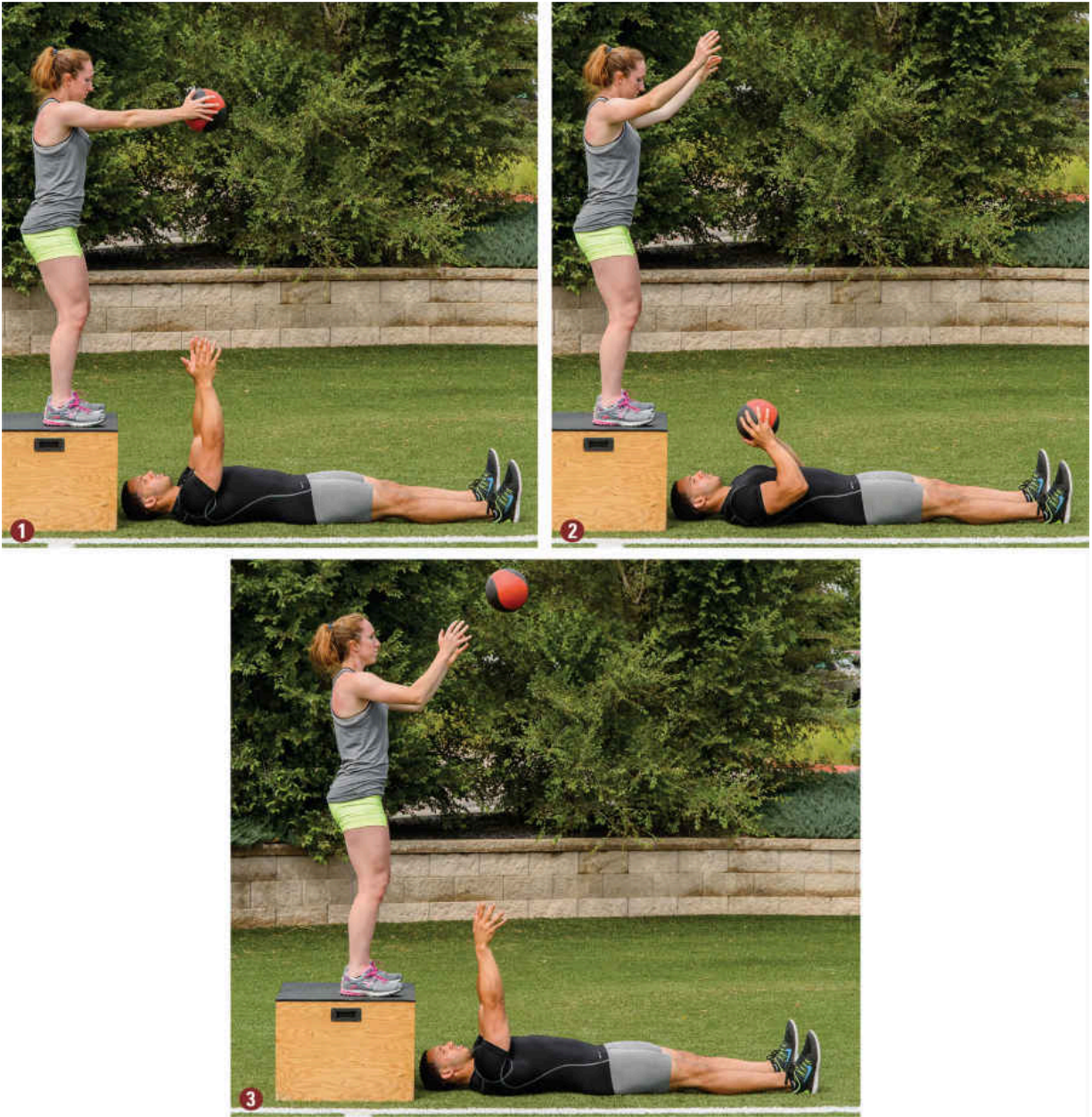
Starting position: Lie supine on the ground with elbows extended and both shoulders in approximately 90 degrees of flexion; the head should be near the base of the box. The partner should be on the box with the medicine ball held above the athlete's arms.

Preparatory movement: None

Arm action: When the partner drops the ball, catch it using both arms and immediately throw the ball back up to the partner.

COMMON ERRORS

- **Catching the ball too high and absorbing the impact force, increasing the amortization phase**
- **Releasing the medicine ball too early, reducing the potential vertical impulse to the ball**



47 HALF-KNEELING OVERHEAD MEDICINE BALL THROW FOR DISTANCE

Intensity level: Low

Direction: Forward

Equipment: Medicine or plyometric ball weighing 5 to 30 pounds (2-14 kg); wall or partner

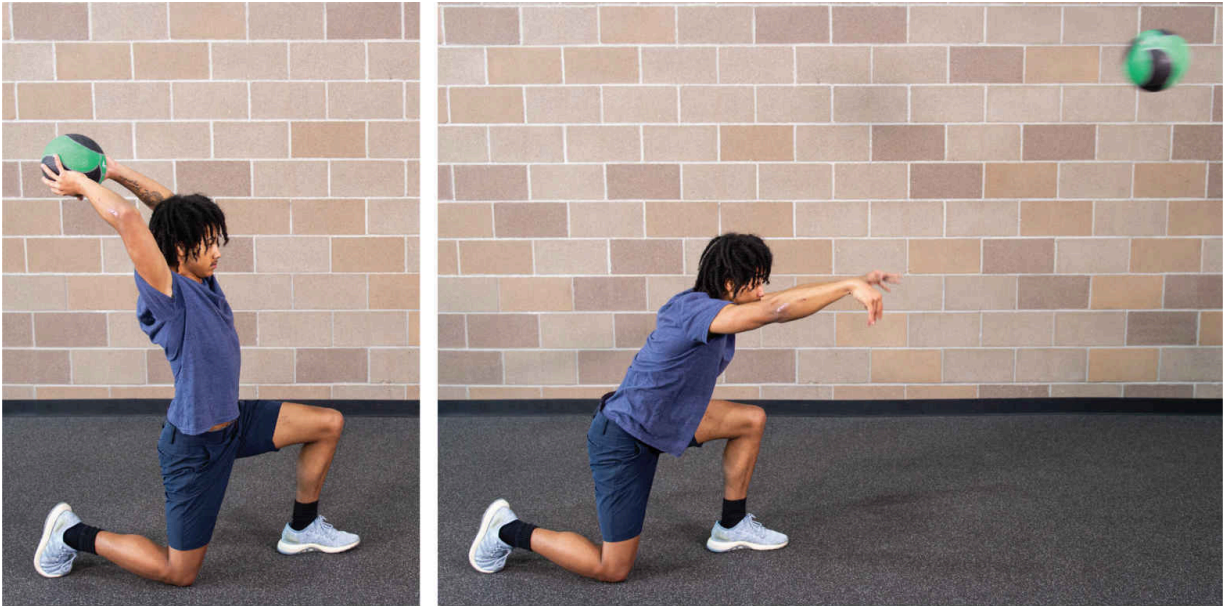
Starting position: Start in a half-kneeling position, with the kneeling knee directly under the hip of the supporting leg and the hip and knee of the stance leg flexed to approximately 90 degrees with the hip, knee, and toe aligned. The trunk should be upright, facing the wall or partner approximately 10 feet (3 m) away (not shown). Raise the ball above the head with the elbows extended.

Preparatory movement: Begin with a countermovement in the trunk as well as the arms, cocking the arms (i.e., moving the arms slightly backward, from the shoulder, before the actual throw).

Arm action: Using both arms, throw the ball to the wall or partner by flexing the trunk and extending the shoulders, maintaining a straight arm position. When the wall or partner returns the ball, return to the starting position, and immediately repeat the movement.

COMMON ERRORS

- **Overextending the trunk in the start position, detrimentally increasing the involvement of the lumbar spine**
- **Producing a cross-body throwing motion (either from not having the ball directly overhead or by finishing with the hands toward one hip); this produces trunk rotation, which may increase the distance but changes the exercise from an isolated sagittal plane movement**



48

SINGLE-ARM REVERSE THROW

Intensity level: Low

Direction: Backward

Equipment: Small medicine or plyometric ball weighing 2 to 30 pounds (1-14 kg); partner

Starting position: Start in a tall-kneeling position, with the kneeling knee directly under the hip and the trunk upright, facing away from a partner who is standing approximately 1 to 2 feet (30-61 cm) behind the athlete. The partner starts with the ball. The athlete starts with the throwing arm raised above the head with the elbow extended.

Preparatory movement: The partner throws the ball over the athlete's head so that its trajectory takes it an arm's length in front of the athlete, traveling toward the opposite knee to the raised arm.

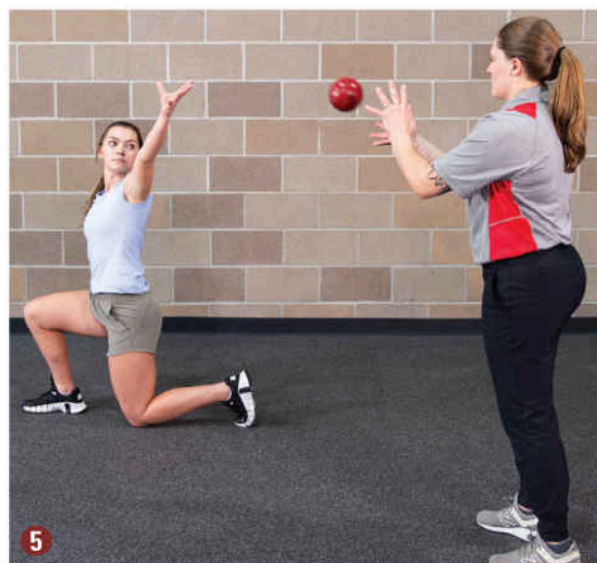
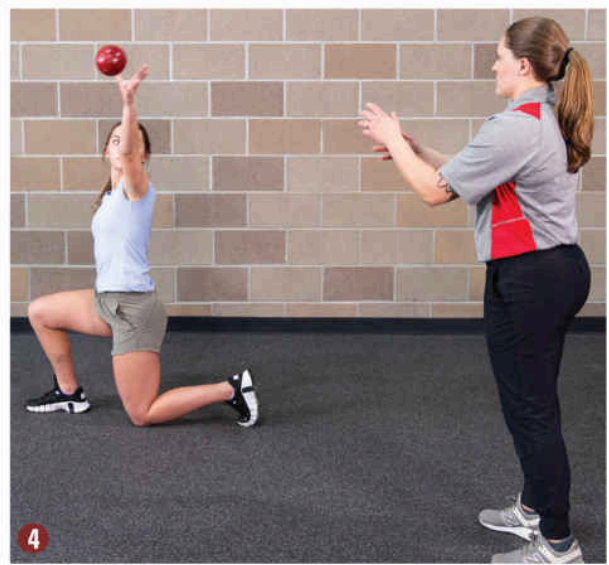
Arm action: The athlete tracks the ball in a diagonal pattern with the throwing arm and catches it with an overhand grip before it hits the ground. Once caught, the athlete keeps the elbow extended while flexing the shoulder in the diagonal pattern, moving to a release

point where the ball is thrown back to the partner when the elbow is fully extended. Immediately repeat the movement.

Note: Because this exercise trains smaller muscle groups involved in scapular control, large increases in implement weight can make a big difference to the intensity of the action.

COMMON ERRORS

- **Using compensatory muscle recruitments to control the shoulder and throw the ball, compromising the objective of the drill**
- **Not catching the ball with sufficient arm extension to challenge the deceleration–reacceleration moments through the rotator cuff as a dynamic shoulder stabilizer**



49

VIKING THROW FOR HEIGHT

Intensity level: Low

Direction: Vertical

Equipment: Medicine or plyometric ball weighing 10 to 50 pounds (5-23 kg)

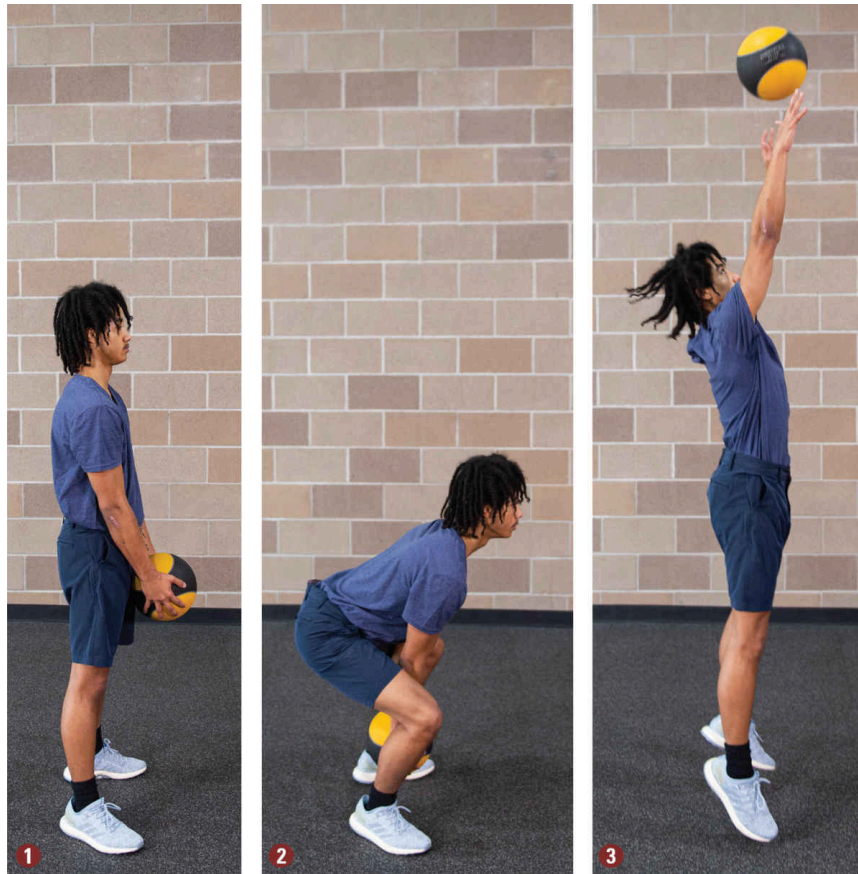
Starting position: Start in an upright stance with the medicine ball held in straight arms in front of the middle of the body.

Preparatory movement: Initiate a countermovement by performing a deep and rapid squat action, keeping the ball held in straight arms.

Arm action: From the base of the countermovement, perform a rapid triple extension of the hips, knees, and ankles to drive the ball vertically. As the body rises, continue the upward momentum of the ball so that it is released and thrown as high as possible at the top of the triple extension.

COMMON ERRORS

- **Not optimizing leg extension and transferring power to the upper body, reducing the impulse to the medicine ball**
- **Releasing the ball too early (before full leg and trunk extension and shoulder flexion), reducing the impulse to the medicine ball**



50 MEDICINE BALL SLAM FROM SINGLE-LEG BALANCE

Intensity level: Low

Direction: Downward

Equipment: Medicine or plyometric ball weighing 10 to 30 pounds (5-14 kg)

Starting position: Start in a single leg Romanian deadlift position, with the trunk parallel to the ground and the hips hinged and level with one leg extended straight behind the body. The stance foot should be flat on the ground with weight distributed toward the heel. The shoulders should be flexed so the straight arms are holding the medicine ball level with the head.

Preparatory movement: Initiate a small countermovement by flexing the shoulders.

Arm action: Keeping the arms straight, slam the ball directly down into the ground, maintaining a stable trunk and hip position. Catch the rebound and repeat the movement, then repeat on the contralateral leg.

Note: With the shoulder position, small increases in load can make a big impact on drill intensity. This drill also can be adapted to a chest-dominant pattern by starting with the ball held at the chest and slammed into the ground from there.

COMMON ERRORS

- **Not achieving a stable and balanced position at the hips to form a stable platform from which to throw the ball**
- **Hyperextending into the lumbar spine to achieve sufficient power to throw the ball to the ground rather than throwing from the shoulder**



Plyometric Push-Ups

51

DEPTH PUSH-UP

Intensity level: Medium

Direction: Vertical

Equipment: Medicine or plyometric ball (or stable implement of similar height)

Starting position: Lie in a push-up position, with the hands on the medicine ball and elbows extended.

Preparatory movement: None

Downward movement: Quickly remove the hands from the medicine ball and drop down. Contact the ground with hands slightly wider than hip-width apart and elbows slightly flexed. Allow the chest to almost touch the medicine ball by letting the elbows flex.

Upward movement: Immediately and explosively push up by fully extending the elbows. Quickly place the palms on the medicine ball and repeat the movement.

Note: When the upper body is at maximal height during the upward movement, the hands should be higher than the medicine ball.

COMMON ERRORS

- **Not achieving sufficient tension in the arms to stop the downward movement of the chest, which collapses onto the medicine ball**
- **Not achieving sufficient bilateral vertical force to land with both hands onto the medicine ball in a balanced fashion**



52

LATERAL TRAVEL PUSH-UP

Intensity level: High

Direction: Lateral

Equipment: Medicine ball or a stable implement of similar height

Starting position: Get into a push-up position, with one hand on the medicine ball (elbow partially flexed) and one hand on the ground, elbow extended (not shown).

Preparatory movement: None

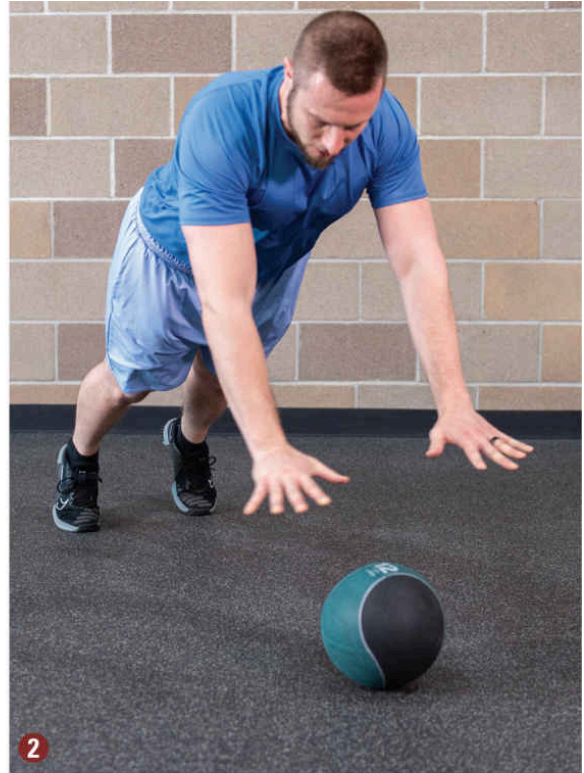
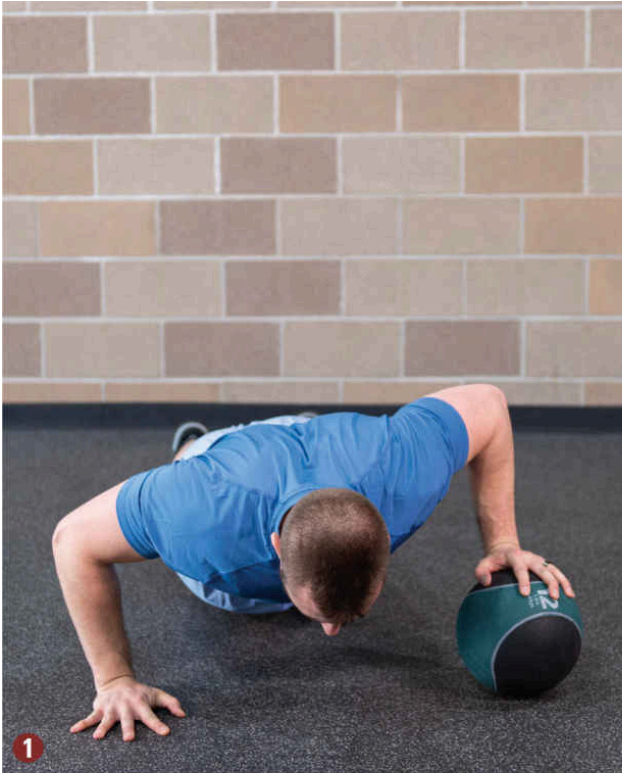
Downward movement: Keeping both hands in place, rapidly drop down into a push-up deep enough so that the chest is level with the top of the medicine ball (or lower).

Drill movement: Immediately and explosively push up and sideways by fully extending the elbows. The trunk should move laterally across the medicine ball while quickly switching hands (the hand that was on the ground now lands on the medicine ball, and vice versa). Immediately on landing, repeat the movement in the opposite direction (alternate hands with each repetition).

Note: When the upper body is at maximal height during the upward movement, both hands should be above the medicine ball.

COMMON ERRORS

- **Leading the vertical movement with the hips rather than pushing through the upper body**
- **Not achieving sufficient vertical height to travel across the ball (which should not move)**



Trunk Plyometrics

53

45-DEGREE SIT-UP

Intensity level: Medium

Equipment: Medicine or plyometric ball weighing 5 to 30 pounds (2-14 kg); a partner

Starting position: Sit on the ground with the trunk at an approximately 45-degree angle. The partner should be in front with the medicine ball.

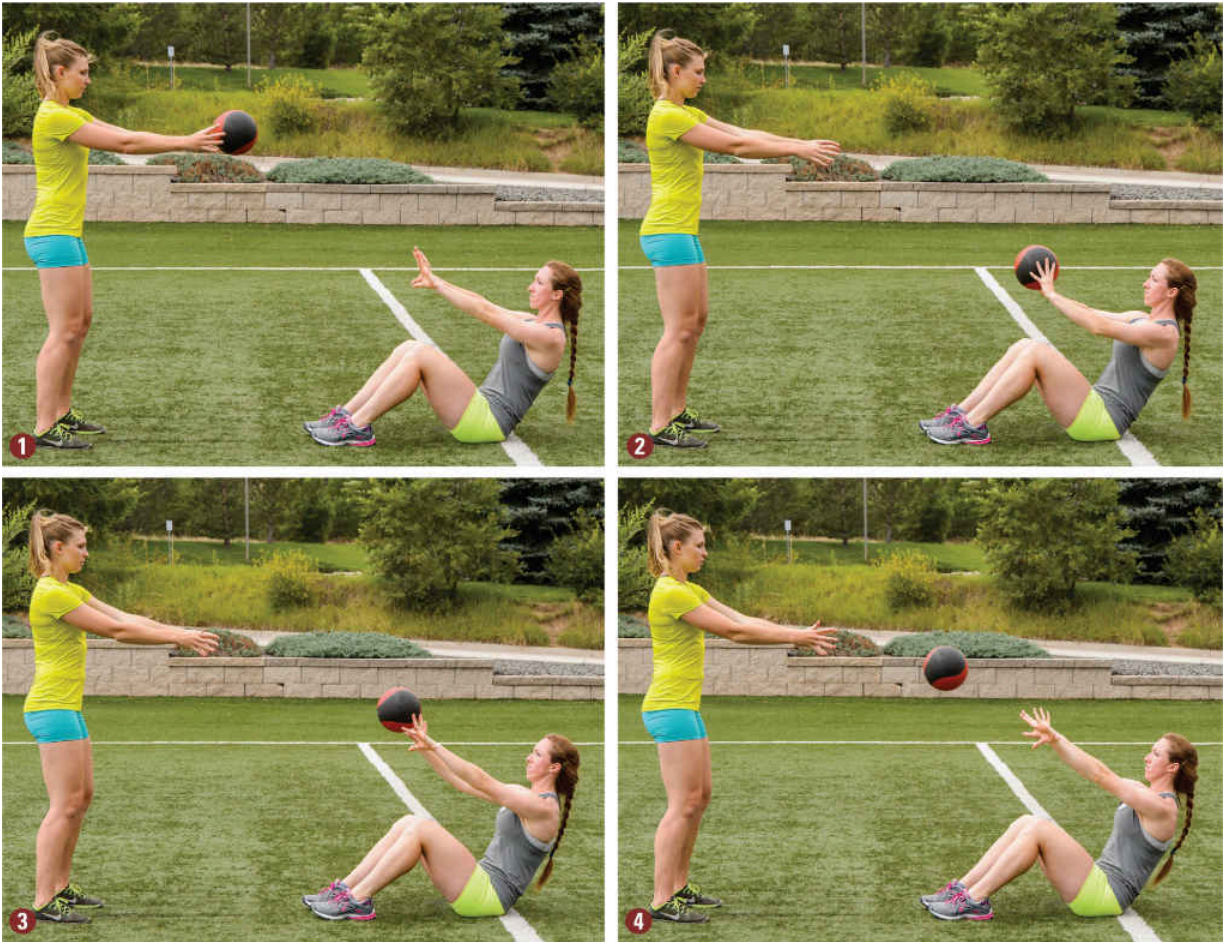
Preparatory movement: The partner throws the ball to the athlete's outstretched hands.

Drill action: Once the partner throws the ball, the athlete catches it using both arms with minimal trunk extension and immediately returns the ball to the partner.

Note: The force used to return the ball to the partner should come predominantly from the abdominal muscles.

COMMON ERRORS

- **Extending at the hips to absorb the force of the ball, increasing the amortization phase and reducing the efficacy of the exercise**
- **Not keeping the foot in contact with the ground, reducing the stable base from which to exert force**



54 SEATED ROTATIONAL MEDICINE BALL THROW

Intensity level: Medium

Equipment: Medicine or plyometric ball weighing 5 to 30 pounds (2-14 kg); a wall or partner

Starting position: Sit on the ground with the trunk at an approximately 45-degree angle. The partner should stand to the side with the medicine ball.

Preparatory movement: The partner throws the ball to the athlete's outstretched hands.

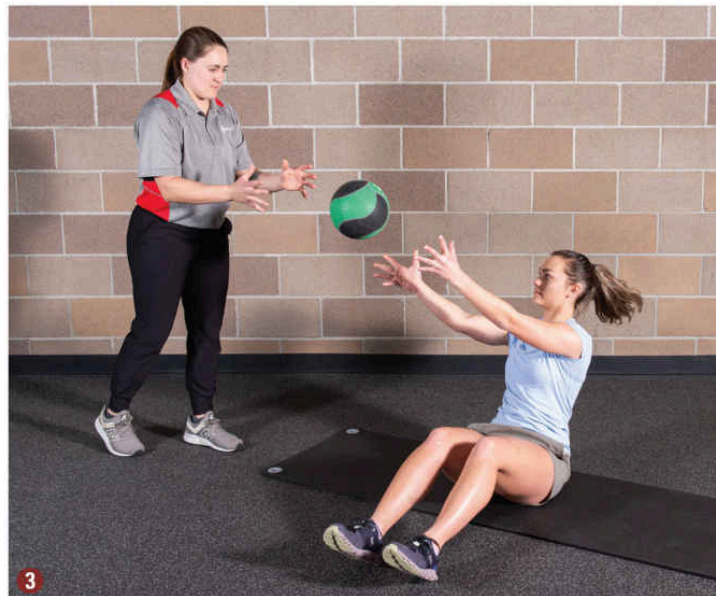
Drill action: Once the partner throws the ball, the athlete should track it across the body and catch it using both arms with minimal

trunk extension. The rotation of the trunk should be decelerated and then rotated in the opposite direction to immediately return the ball to the partner.

Note: The force used to return the ball to the partner should come predominantly from the abdominal muscles and trunk rotation.

COMMON ERRORS

- **Not rotating sufficiently to optimize the SSC within the trunk musculature**
- **Not keeping the foot in contact with the ground, reducing the stable base from which to exert force**



55 LEG PUSHDOWN

Intensity level: High

Equipment: A partner

Starting position: Lie on the ground with slightly flexed knees and the hips flexed to 90 degrees (not shown). The hands should hold the partner's ankles for stability (shown).

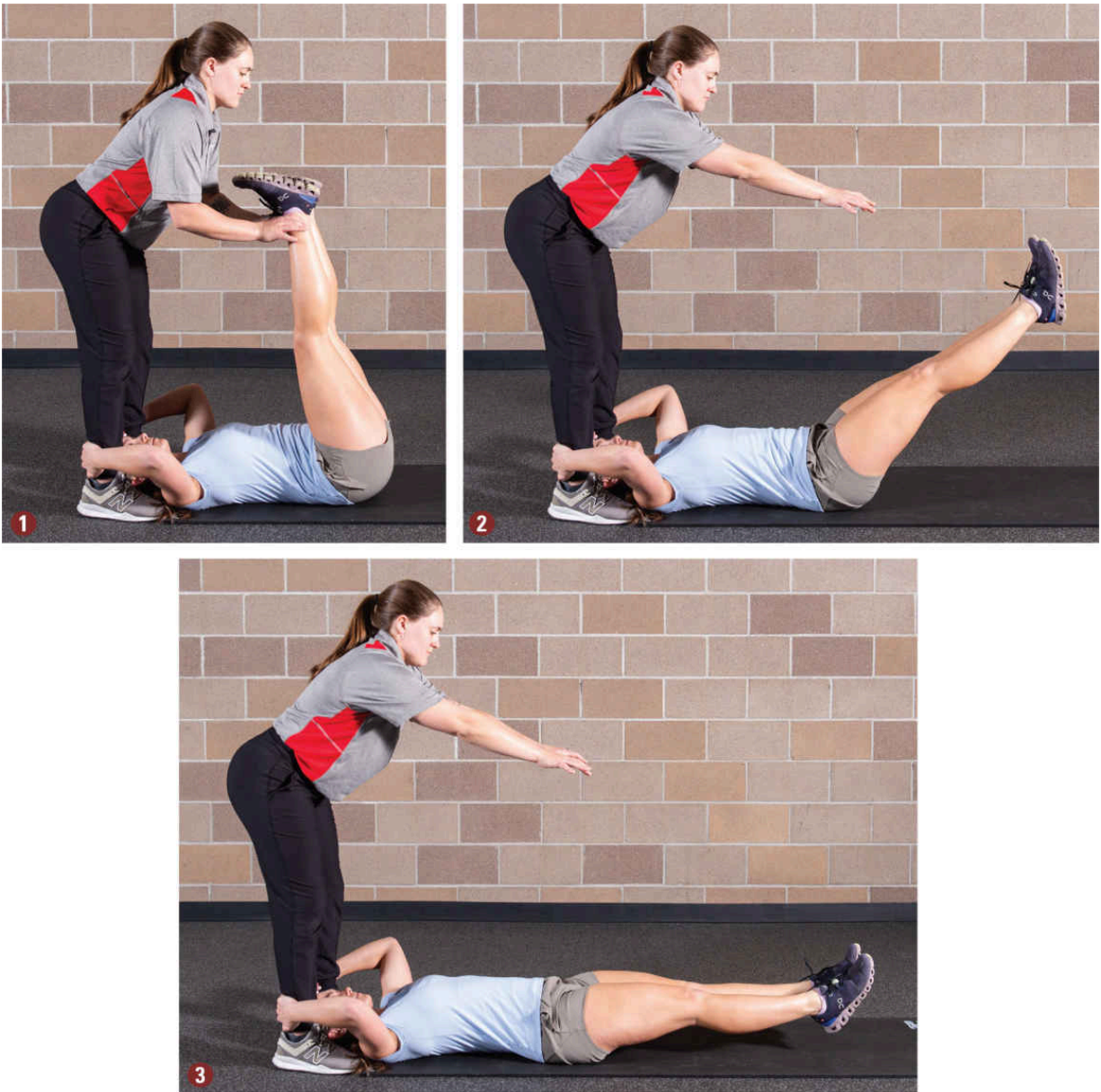
Preparatory movement: Brace the core. Breathe in and pull the ribs down toward the hips to tighten the core musculature to prevent the lumbar spine from going into forced extension during the movement.

Drill action: The athlete raises the legs toward the partner's outstretched hands, and the partner briefly grasps the athlete's lower legs before forcefully pushing the athlete's legs down toward the ground. Before the legs reach the ground, the athlete decelerates and ultimately stops the downward movement before flexing the hips to return to the starting position to reset for the next repetition. Immediately repeat the movement.

Note: Intensity may be increased by using greater force of the downward push or by altering the direction of the push so that the leg action is not purely in the sagittal plane.

COMMON ERRORS

- **Lowering the legs without waiting for the partner to push them to the ground, reducing the eccentric load of the exercise**
- **Not controlling the lumbopelvic position and extending the spine as the legs are lowered, increasing the pressure on the lower back musculature**



KEY TERMS

active flat foot
 amortization phase
 ascending complex training
 back squat equivalent strength
 balance
 ballistic exercise
 bound

box jump
complex training
concentric phase
contrast training
coupling phase
depth jump
descending sets
dynamic valgus
eccentric phase
foot contact
hop
jump
muscle spindle
negative foot speed
parallel elastic component (PEC)
plyometric exercise
potentiation
power
progressive overload
rebound exercise
series elastic component (SEC)
slam
stretch reflex
stretch reflex latency
stretch-shortening cycle (SSC)
throw

STUDY QUESTIONS

1. Stored elastic energy is used to increase force production during which phase of plyometric exercises?
 - a. eccentric
 - b. isometric
 - c. amortization
 - d. concentric